INTERNATIONAL FOOTBALL: A 150- Year Data Journey (1872-2026)

Exploratory Data Analysis of 47,000+ Matches (1872–2024)

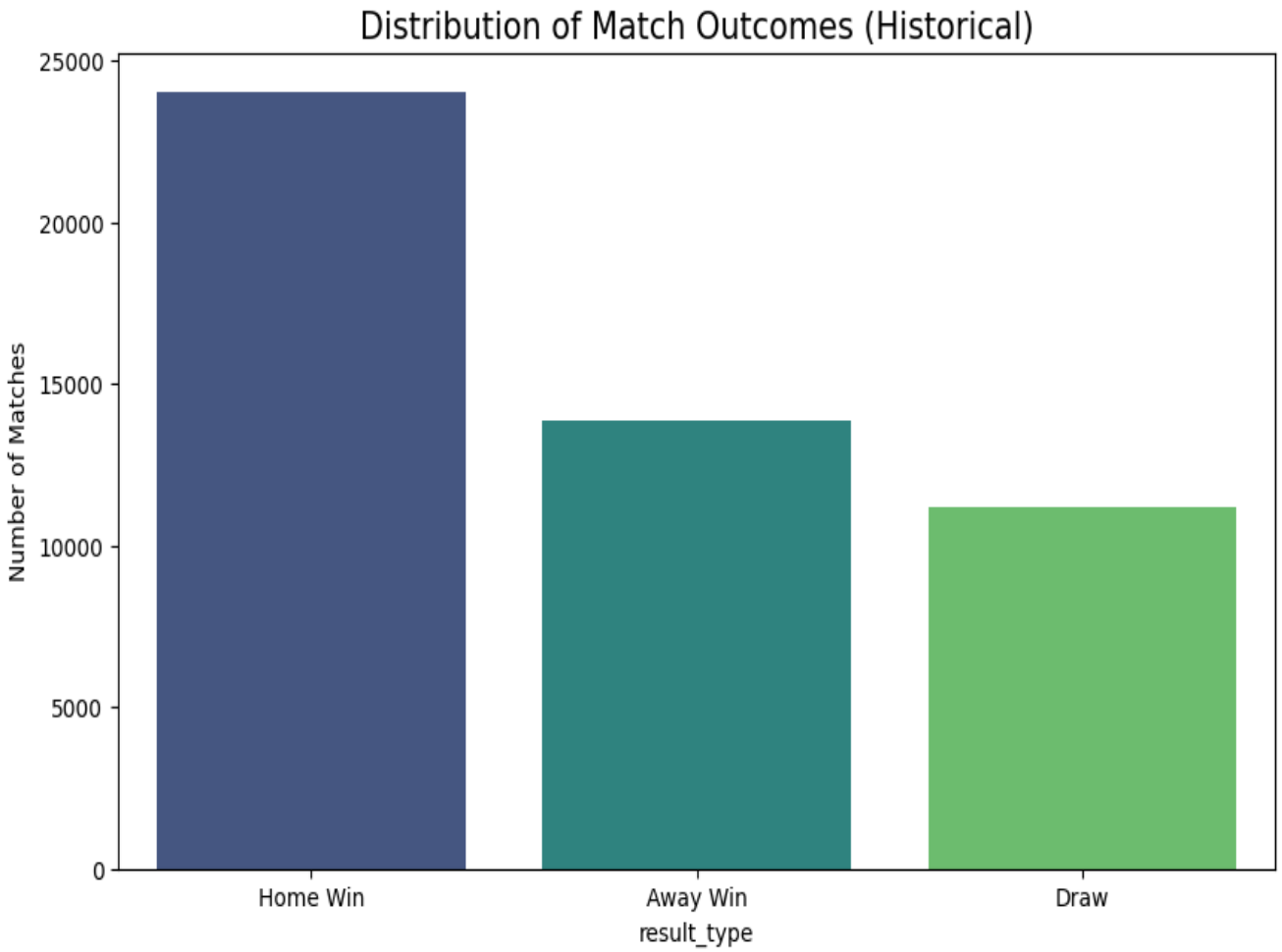
- **Presented by:** Jahnvi kumai
- **Tools Used:** Python (Pandas), Matplotlib, Seaborn

Data Overview & Cleaning

- **The Dataset:** Analyzing four core files(results, shootouts, goalscorers and formal_name) to uncover historical trends

- **Key Goals:**
  - Understand the growth of international football over 15 decades.
  - Quantify the statistical significance of "Home Field Advantage."
  - Resolve geopolitical data inconsistencies (mapping historical nations to modern successors)

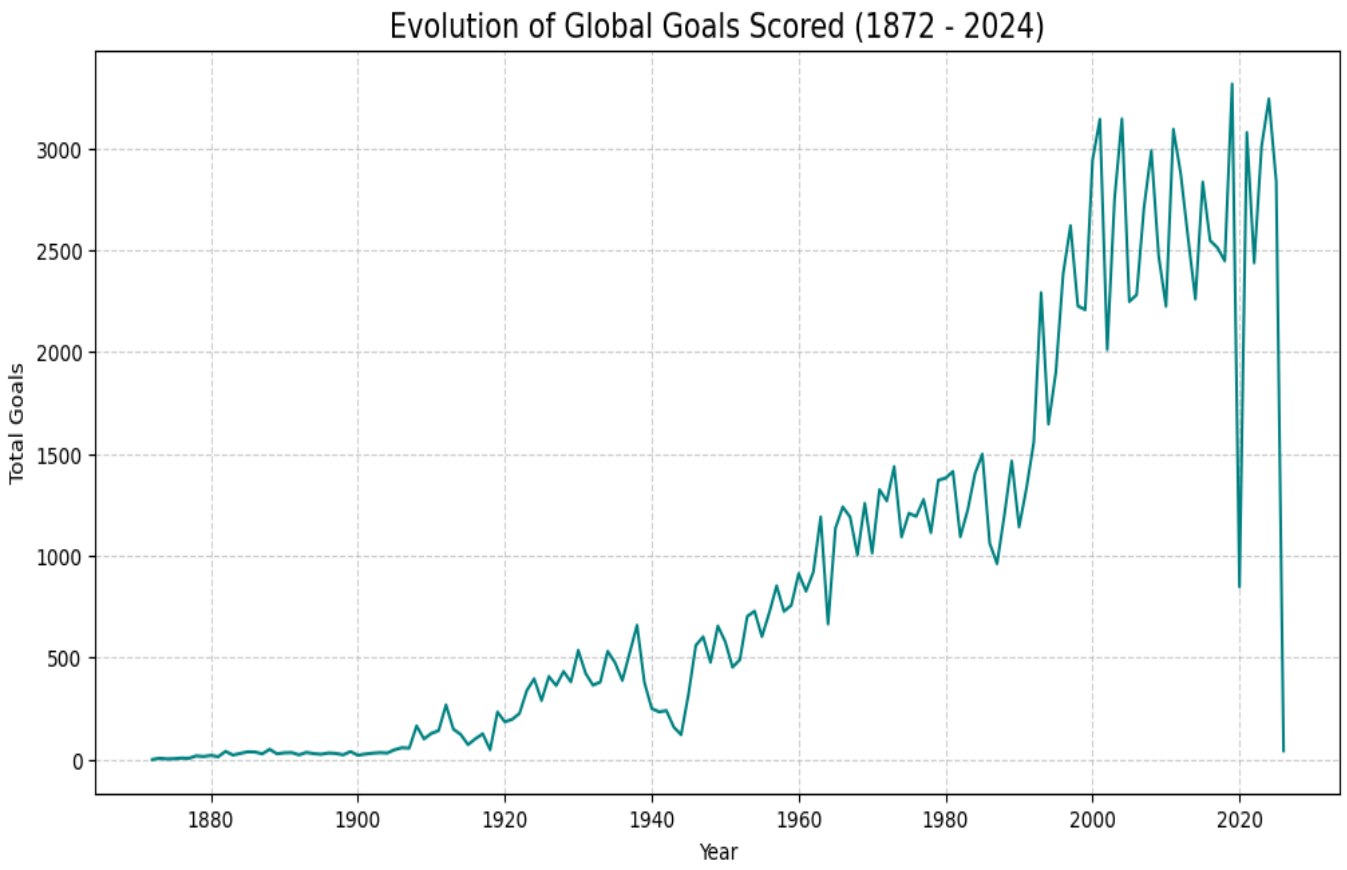
Visualizing the Home Field Advantage (HFA)



> Home teams consistently win more often than away teams

> Historically, Home Wins account for nearly **50%** of all results. This suggests that crowd support, familiar environments, and lack of travel fatigue are statistically significant factors in international success

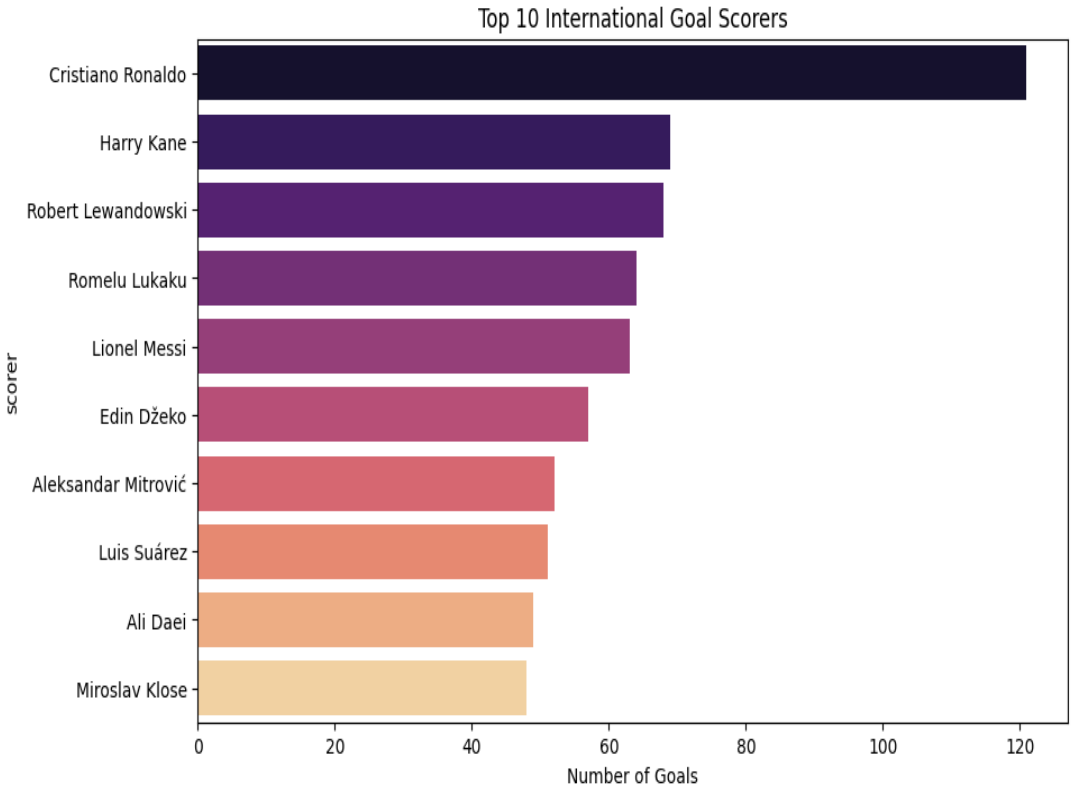
Visualizing the Growth of the Sport



> A massive explosion in match frequency occurred post-1970, marking the "Modern Professional Era

> Spikes in goals often align with World Cup years, showing increased global participation and the expansion of FIFA member nations.

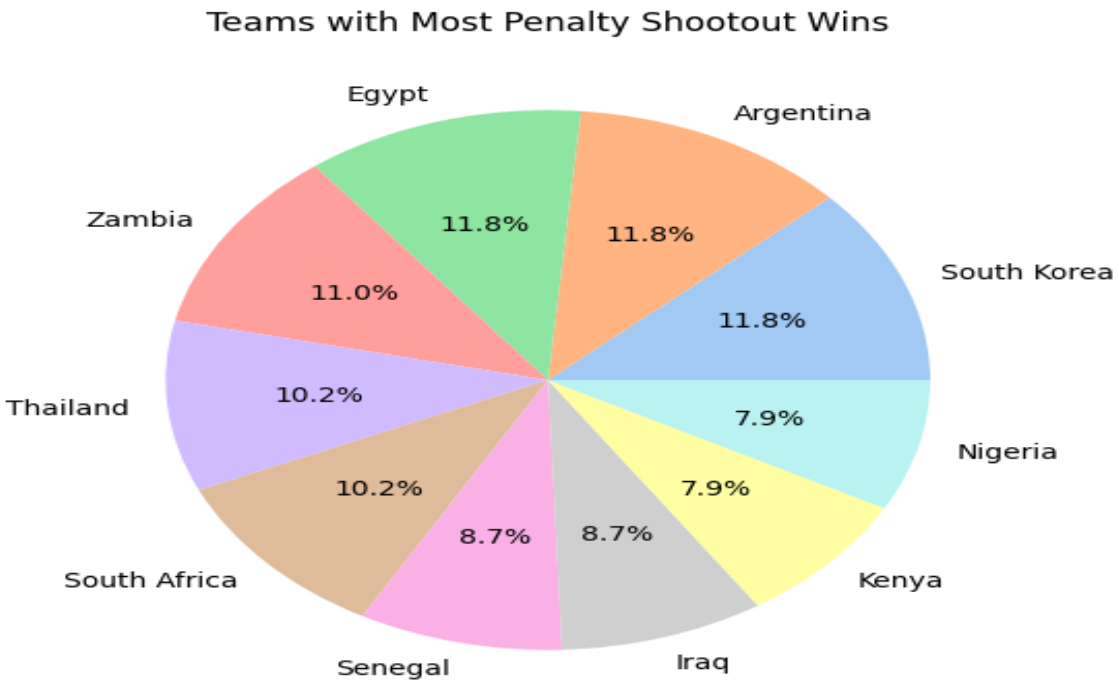
The Legends (Top Scorers)



> By merging the results and goalscorers data, we can see that players like [Player Name] have dominated. This highlights the longevity required to stay at the top of international sports.

> Cristiano Ronaldo is the most consistent goal machines.

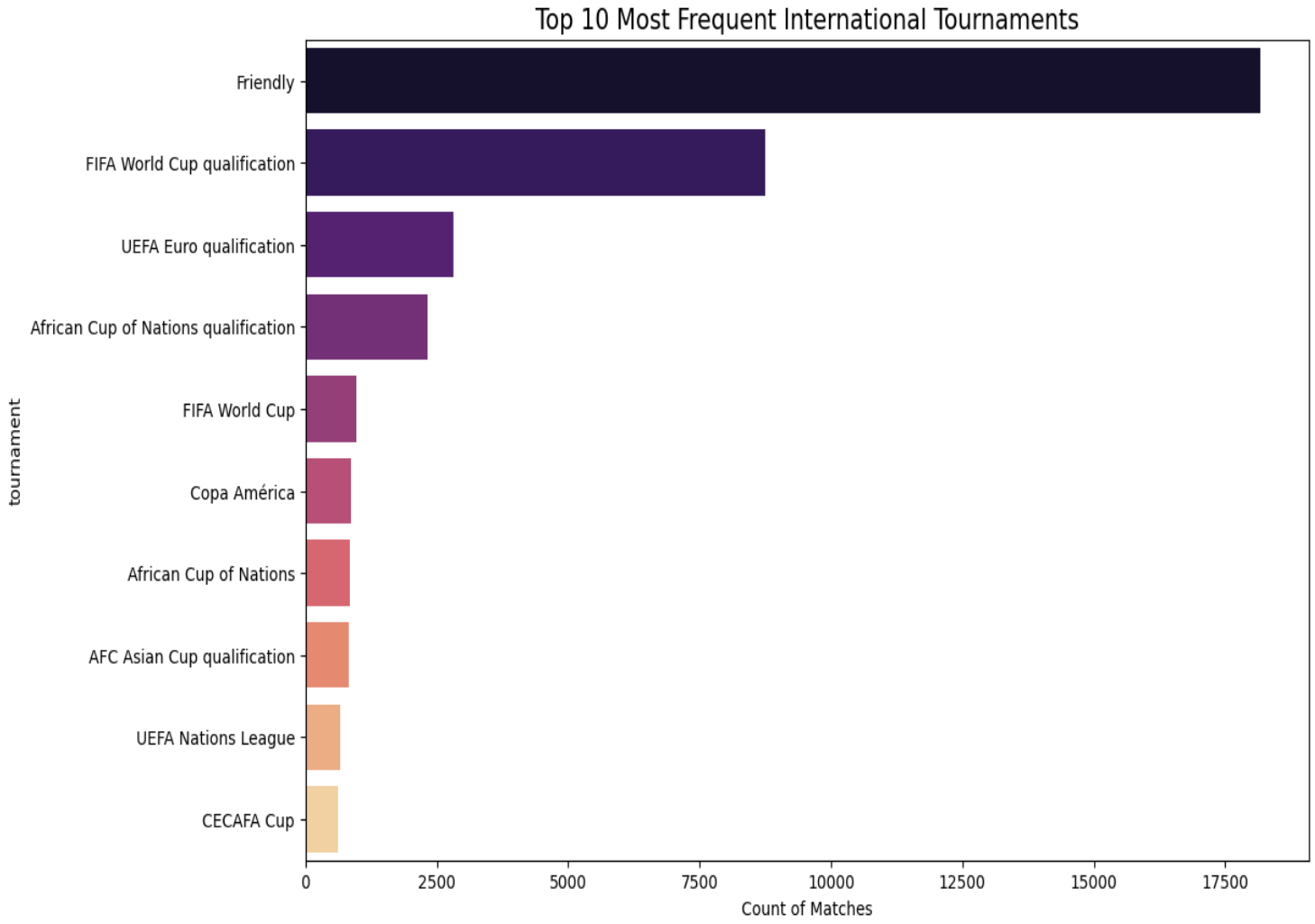
Pressure Situations (Shootouts)



> Penalty shootouts are often called a 'lottery,' but the data shows that certain nations, like [Country], have a significantly higher success rate than others."

> **Key findings-** Which countries handle the pressure of penalty kicks the best?

Tournament Diversity



> While the FIFA World Cup is the most prestigious, **Friendly Matches** dominate the dataset

> Over 35% of the matches in this 150-year history are "Friendly" games. While they provide volume, they often lack the high-stakes intensity of competitive play.

Project Conclusion

Through this comprehensive analysis of over **150 years of international football**, it is evident that data serves as a powerful bridge between historical geopolitical shifts and modern athletic performance. Implementing a robust **data-Cleaning and preparation** this study ensured a continuous and accurate timeline of global dominance. The findings statistically validate the enduring power of **Home-Field Advantage.** Ultimately, this project demonstrates that while the game on the pitch has evolved in speed ,the underlying statistical patterns provide a reliable foundation for predictive modelling and strategic insights into the future of the world's most popular sport.

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